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QCBench: Evaluating Large Language Models on Domain-Specific Quantitative Chemistry

Jiaqing Xie,[▽] Weida Wang,[▽] Ben Gao,[▽] Zhuo Yang, Haiyuan Wan, Shufei Zhang, Tianfan Fu, and Yuqiang Li*

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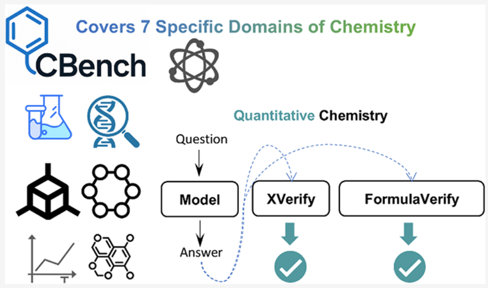
ABSTRACT: Quantitative chemistry is central to modern chemical research, yet the ability of large language models (LLMs) to perform its rigorous, step-by-step calculations remains underexplored. To fill this gap, we propose QCBench, a **Q**uantitative **C**hemistry oriented benchmark comprising 350 computational chemistry problems across 7 chemistry subfields, which contains analytical chemistry, bio-organic chemistry, general chemistry, inorganic chemistry, physical chemistry, polymer chemistry and quantum chemistry. To systematically evaluate the mathematical reasoning abilities of large language models (LLMs), they are categorized into three tiers: easy, medium, and difficult. Each problem, rooted in realistic chemical scenarios, is structured to prevent heuristic shortcuts and demand explicit numerical reasoning. QCBench enables fine-grained diagnosis of computational weaknesses, reveals model-specific limitations across difficulty levels, and lays the groundwork for future improvements such as domain-adaptive fine-tuning or multimodal integration. Evaluations on 24 LLMs demonstrate a consistent performance degradation with increasing task complexity, highlighting the current gap between language fluency and scientific computation accuracy. Code for QCBench is available at https://github.com/jiaqingxie/QCBench.



■ INTRODUCTION

Numerous benchmark data sets have been proposed to evaluate the problem-solving capabilities of AI models across scientific disciplines, including mathematics,[1−3] physics,[4−6] and chemistry.[7−9] In the domains of mathematics and physics, benchmark problems predominantly focus on theorem proving and numerical computation. However, analytical or comprehension-based problems, which require deeper semantic understanding, are underrepresented in mainstream math and physics data sets.[4,5,10−12] In contrast, chemistry benchmarks often involve complex reasoning tasks that rely heavily on structural interpretation and image-based analysis. For instance, many problems require interpreting H NMR spectra,[13] evaluating molecular properties from structural diagrams,[14] or performing reaction synthesis prediction.[8,9,15] Despite this diversity, current chemistry benchmarks remain limited in their breadth and depth. As shown in Figure 1, only a small fraction of tasks in general-purpose benchmarks such as MMLU[16] and OlympicArena[17] are computational in nature, constituting merely 10 and 14 problems, respectively, while over 80% are categorized as nonquantitative.

A notable limitation of existing chemistry benchmarks lies in the incomplete and often ambiguous categorization of problem domains. For instance, SciBench[7] includes tasks primarily drawn from *Quantum Chemistry* and *Physical Chemistry*, while ChemBench[9] encompasses seven chemistry domains. However, upon closer inspection, the *Analytical Chemistry* problems in ChemBench predominantly revolve around tasks such as determining the number of hydrogen atoms from ^{1}H NMR spectra.[13] We claim that such tasks should be categorized as **predictive** *rather than* **quantitative** (Figure 2). Truly quantitative problems in scientific domains involve explicit computational reasoning, typically requiring the application of formulas, step-by-step calculations (e.g., equations, numerical approximations), and yielding numerical outputs, often as floating-point values.[19] Furthermore, tasks such as chemical equation balancing, although appearing procedural, do not constitute genuine quantitative reasoning. They primarily test conceptual understanding of chemical principles, such as the conservation of mass, rather than mathematical computation. Similarly, predicting the number of structural isomers or determining electron counts in molecules involves rule-based inference or pattern recognition, and should not be conflated with quantitative problem solving. This misclassification leads to a second drawback: many so-called computational tasks in existing benchmarks do not reflect the depth of quantitative reasoning expected in math or physics domains. As shown in

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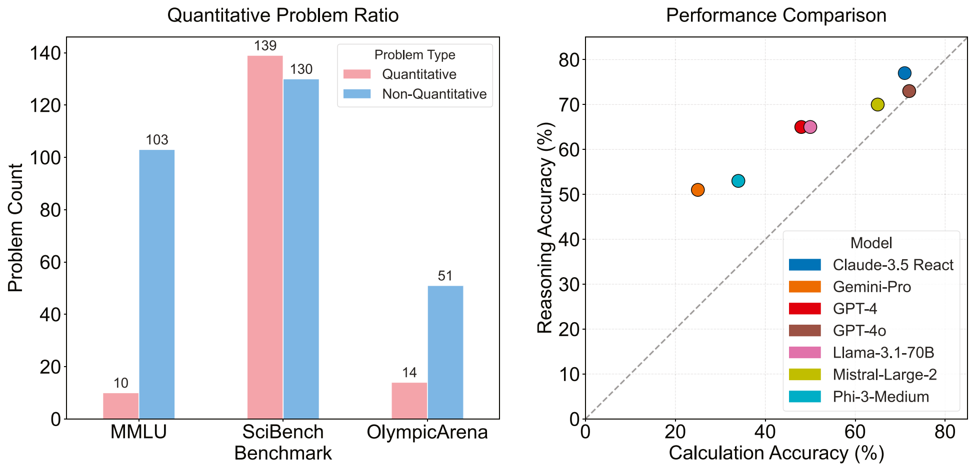

Figure 1. Quantitative questions ratio is relatively small in some chemistry benchmarks. Also, performance is relatively high for reasoning tasks as observed by ChemBench,[9] which motivates us to curate a pure computing benchmark. Original figure created by the authors; not previously published.

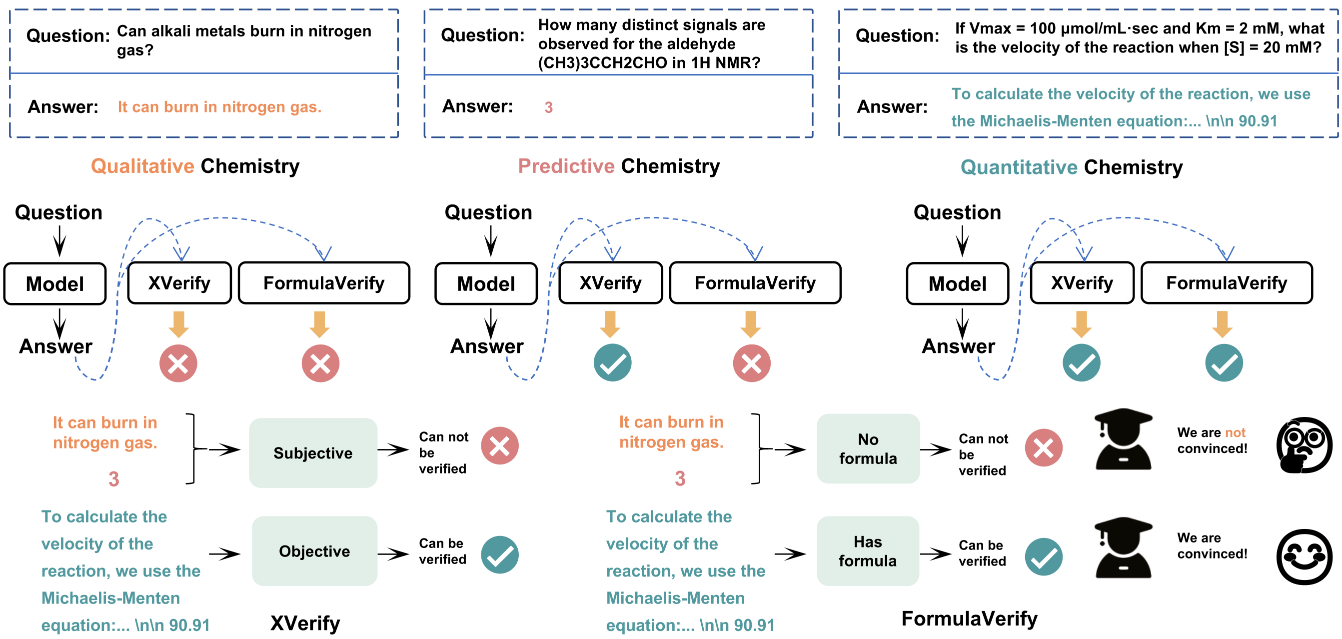

Figure 2. *Qualitative chemistry* versus *predictive chemistry* versus *quantitative chemistry*. We define *quantitative* chemistry problems as those requiring objective, numerical answers that can be automatically evaluated by tools such as xVerify.[18] In contrast, qualitative problems involve textual explanations or conceptual reasoning without definitive ground-truth answers. *Predictive* chemistry includes simple counting tasks, such as tallying hydrogen atoms, or interpreting reaction equations, as they lack formal computational steps and do not require formula-based reasoning.

the right panel of Figure 1, this discrepancy is further reflected in model performance. Current state-of-the-art models demonstrate stronger reasoning and pattern recognition capabilities than true computational competence in chemistry tasks, underscoring the need for more rigorous quantitative benchmarks.

Another challenge in current chemistry benchmarks is the lack of standardized difficulty calibration. Existing benchmarks span a wide range of problem complexities. Some focus exclusively on Olympiad-level tasks designed for high-achieving students in competitive settings,[6,17] while others include relatively simple high school or college-entry questions,[7,16] which pose little challenge to experienced students such as IChO participants. A more systematic categorization of problem difficulty is essential for meaningful evaluation across ability levels. Furthermore, evaluating quantitative chemistry problems introduces a critical methodological tension in answer verification. On one hand, the objective rigor offered by strict verification systems, such as xVerify,[18] is essential. By relying on exact matching, these tools are vital for ensuring computational soundness and penalizing answers derived from flawed reasoning. On the other hand, this deterministic approach can be fundamentally misaligned with chemical practice. Unlike pure mathematics, quantitative chemistry inherently involves numerical tolerances, significant figures, and rounding conventions derived from experimental data. A verification process that is too rigid risks unfairly penalizing models that reason correctly but yield a numerically equivalent, yet nonidentical, output. It motivates a core aspect of our work: not merely to choose one method over the other, but to employ both strict and tolerance-based verification. By analyzing the performance gap between these two approaches,

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we can uncover deeper insights into a model's underlying reasoning and self-correction capabilities.

Main Contributions. In this work, we make the following key contributions:

(1) We design the benchmark called **QCBench**, which is specifically focused on **Q**uantitative **C**hemistry problems, such as tasks requiring explicit numerical computation, and comprising 350 problems across seven major subfields with three difficulty levels. The benchmark includes both expert-curated problems and samples adapted from existing data sets. All problems can be formalized and solved using symbolic or neural reasoning methods.

(2) We provide the first systematic analysis of their capabilities on domain-specific quantitative chemistry. Our findings reveal a critical gap between linguistic fluency and computational accuracy, a consistent performance decline with increasing problem difficulty, and distinct model-specific strengths and weaknesses across different chemical domains.

(3) We introduce a methodological insight by analyzing the performance gap between strict (e.g., xVerify) and tolerance-based verification. We demonstrate that for advanced models, this gap serves not as a measure of formatting error, but as a powerful indicator of sophisticated self-correction capabilities, transforming the verification process itself into a diagnostic tool for advanced reasoning.

RELATED WORKS

Chemistry Benchmarks. We categorize existing chemistry benchmarks into two types based on input modality: single-modality benchmarks, which present problems purely in textual form, and multimodality benchmarks, which incorporate both textual and visual inputs, such as molecular structures or crystallographic images. Representative single-modal benchmarks include ChemBench,[9] ChemLLMBench,[8] MMLU,[16] SciBench,[7] SciCode,[23] JEEBench,[21] and OlympicArena.[6] In contrast, multimodal chemistry benchmarks such as ScienceQA,[22] OlympiadBench,[6] MaCBench,[24] ChemTable,[25] MolPuzzle,[14] Emma,[26] and ScemQA[27] often emphasize visual understanding and concept recognition tasks. While multimodal benchmarks have grown in number and scope, they are predominantly centered on conceptual or interpretive tasks, which fall outside the scope of our study. In this work, we focus on single-modality benchmarks and further analyze their coverage of chemistry subfields. Specifically, we classify benchmark questions into seven vertical domains of chemistry and summarize this categorization in Table 1. Our analysis reveals that most existing benchmarks focus narrowly on one or two types of computational problems, lacking both domain coverage and problem diversity. This motivates the development of our benchmark, which aims to address these limitations through broader coverage and a focus on quantitative problem-solving.

LLMs for Solving Chemistry Problems. Recent advances in large language models (LLMs) have enabled them to tackle a broad range of chemistry problems, which can be broadly categorized into three types: qualitative, predictive, and quantitative. *Qualitative* chemistry problems require reasoning or structural understanding, such as assigning functional groups in spectra, interpreting reaction mechanisms, or deducing morphology−property relationships. GPT-4 has shown strong performance in such tasks, although it still struggles with complex reasoning in stereochemistry or crystallography.[7,16] *Predictive* chemistry problems involve direct property or reaction outcome predictions, where answers are derived from learned associations rather than explicit computation. These include predicting boiling points, reactivity, or products of simple reactions, often without requiring intermediate steps. Closed-source models like GPT-4 and GPT-4o have achieved notable accuracy on such tasks.[8] A recent study has shown that o1 is the best model across calculation, knowledge inference and reasoning.[9] Despite recent progress, most benchmarks report only aggregate accuracy, without differentiating model performance across these problem types or difficulty levels. This limits our understanding of LLM strengths and weaknesses in sub-chemistry domains, and motivates the development of more structured evaluations.

Table 1. Comparison of Chemistry Benchmarks by Subcategory Coverage

benchmark	ref	subcategories (only quantitative problems counted)						
		PhC	IC	QC	AC	BOC	GC	PoC
SciEval	20							
OlympicArena	17	√	√		√		√	
SciBench	7	√		√				
JEEBench	21	√	√					
ChemBench	9	√	√		√			√
OlympiadBench	6							
ScienceQA	22							
SciCode	23			√	√			
QCBench	ours	√	√	√	√	√	√	√

CONSTRUCTION OF QCBENCH

Quantitative chemistry problems require numerical reasoning grounded in formulas, constants, and multistep derivations, such as computing Gibbs free energy or equilibrium concentrations. These tasks remain difficult for LLMs, especially in *Quantum* and *Physical Chemistry*, as shown by poor performance on SciBench[7] and college-level MMLU.[16] To systematically assess this capability, we propose QCBench (Figure 3), the **Q**uantitative **C**hemistry **Bench**mark, specifically designed for quantitative problems defined in the field of chemistry. It consists of three main components: data filtering and curation, large language model evaluation, and answer verification.

Subset QCBench-84 vs QCBench-350. We observe that QCBench is an imbalanced data set, where a large proportion of problems fall into the Physical Chemistry domain. To enable a fairer comparison across domains, we curate a balanced subset, QCBench-84 (Table 3), from the original QCBench (QCBench-350) (Table 2). Specifically, we randomly sample 12 problems for each domain, with a fixed random seed to ensure reproducibility and to avoid any subjective bias in selection. While QCBench-350 serves as the full-scale benchmark for comprehensive evaluation, QCBench-84 provides a smaller and more balanced testbed (Table 3).

Assessing LLMs on QCBench. As shown in Tables 4 and 5, we evaluated a total of 24 large language models on QCBench, including 11 proprietary models, such as Claude 3.5 Sonnet, GPT-4o,[28] and Gemini 2.5 pro[29] and 13 open-source

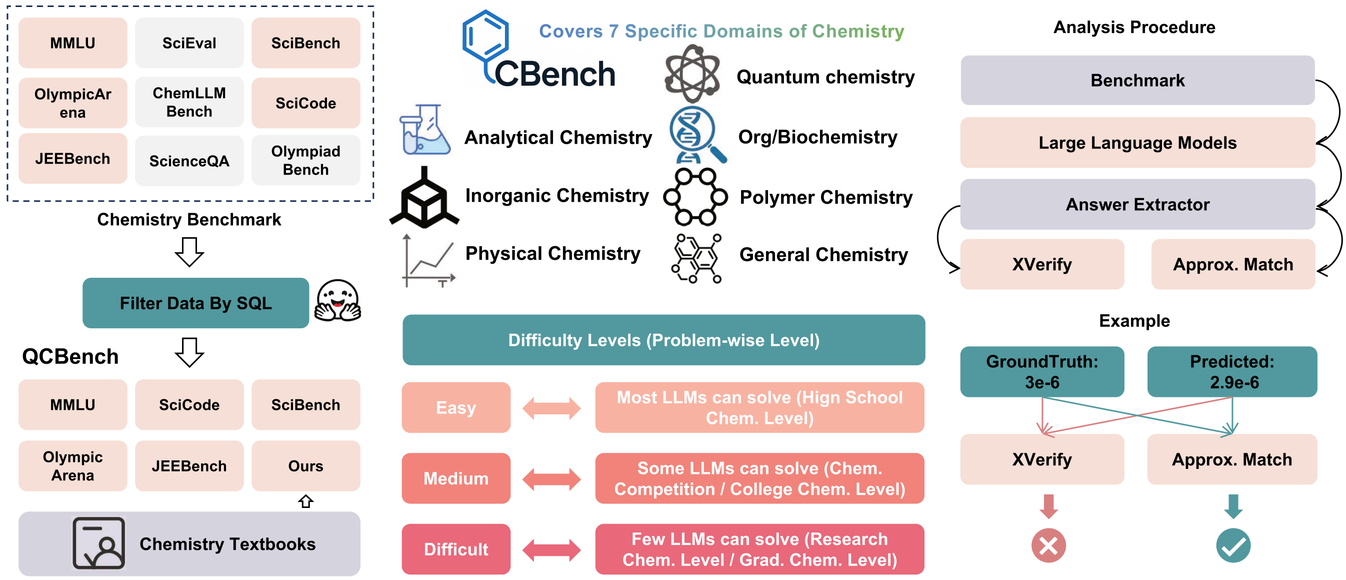

Figure 3. Framework of QCBench.

Table 2. Distribution of Question Difficulties Across Chemistry Subfields in Full QCBench (350 Questions) by Question Length

class	abbrev.	easy	medium	difficult
Analytical Chemistry	AC	6	4	15
Bio/Organic Chemistry	BOC	2	9	14
General Chemistry	GC	2	4	10
Inorganic Chemistry	IC	11	16	19
Physical Chemistry	PhC	29	74	84
Polymer Chemistry	PoC	1	8	3
Quantum Chemistry	QC	19	13	7
total		70	128	152

Table 3. Distribution of Question Difficulties Across Chemistry Subfields in Subset QCBench (84 Questions) by Question Length

class	abbrev.	easy	medium	difficult
Analytical Chemistry	AC	2	2	8
Bio/Organic Chemistry	BOC	1	5	6
General Chemistry	GC	2	3	7
Inorganic Chemistry	IC	1	5	6
Physical Chemistry	PhC	1	3	8
Polymer Chemistry	PoC	1	8	3
Quantum Chemistry	QC	5	5	2
total		13	31	40

models, such as DeepSeek-R1,[30] and Qwen3−235B.[31] These models were selected to cover a broad spectrum of capabilities, licensing terms, and training scales, ensuring a comprehensive assessment of current LLM performance in quantitative chemistry tasks.

Answer Verification. We adopt two complementary approaches for answer verification: (i) a strict verifier based on xVerify, and (ii) a custom tolerance-based pipeline tailored for chemistry problems. **(i) xVerify-based strict verifier.** For tasks with deterministic answers, common in mathematics and physics, xVerify[18] serves as a reliable tool. Specifically, we use the scalable version of xVerify-0.5B−I to evaluate the output of each LLM. **(ii) Tolerance-based evaluation pipeline.** Unlike mathematics and physics, in quantitative chemistry, slight approximations are often acceptable due to the nature of experimental data and rounding conventions. To accommodate this, we design a relaxed evaluation pipeline that tolerates small relative errors while maintaining fairness. Specifically, we provide two verification modes:

- Basic mode: Supports exact numeric formats including integers, floating-point numbers, scientific notation (e.g., $a \times 10^{-b}$, $a * e{-}b$), fractions (a/b), and mixed fractions ($5a/b$). A default relative error tolerance of 1×10^{-6} is applied.
- Pro mode: Incorporates decimal precision matching by comparing the number of decimal places in the predicted answer with that of the ground truth, allowing for context-aware flexibility.

For fair comparison, we adopt the pro mode by default. The ablation study on tolerance error under the basic mode is discussed later. Furthermore, we conducted an ablation study to assess the impact of different temperature settings on performance stability. The results, detailed in the Supporting Information (Section: Different Temperatures), confirm that our main findings are robust to this hyperparameter.

■ RESULTS AND ANALYSIS

Overall: Discriminative Power of Questions. Our test results (Tables 4 and 5) show that no current large language model achieves state-of-the-art performance across all subfields. For instance, some models like **GPT-oss-120B** may excel in *Biochemistry*, while others like **QWQ-32B** perform better in *Polymer Chemistry*. Furthermore, the results indicate that more parameters do not necessarily correlate with better performance. In fact, for large models not explicitly designed or trained for chemistry-related computations, the effects of scaling may not be apparent at all.

QCBench-84 vs QCBench-350. A comparison of the results from the balanced QCBench-84 (Table 4) and the full QCBench-350 (Table 5) reveals a high degree of consistency in the overall performance hierarchy of the models. The relative rankings of the models remain largely stable; for instance, **o3** is the top-performing model in both data sets, and

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Table 4. Approximation and Exact Matching (by xVerify) Accuracies (%, ↑) of Models Across 7 Chemistry Subfields, with Columns for Approximate (Appr.) and xVerify Results[a]

model	Analytical		Biochemistry		General		Inorganic		Physical		Polymer		Quantum		avg. (perf.)	
	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify
closed-source models																
Claude-3.5-Sonnet	$19.4_{4.8}$	$13.9_{9.6}$	$30.6_{4.8}$	$33.3_{0.0}$	$27.8_{4.8}$	$27.8_{4.8}$	$50.0_{0.0}$	$50.0_{0.0}$	$33.3_{0.0}$	$33.3_{0.0}$	$\underline{33.3_{14.4}}$	$27.8_{9.6}$	$25.0_{8.3}$	$22.2_{9.6}$	$31.3_{4.5}$	$29.2_{3.7}$
Claude-4-Sonnet	$37.5_{5.9}$	$37.5_{5.9}$	$\underline{41.7_{0.0}}$	$\underline{41.7_{0.0}}$	$45.8_{5.9}$	$\underline{54.2_{5.9}}$	$58.3_{11.8}$	$58.3_{11.8}$	$\mathbf{41.7_{0.0}}$	$37.5_{5.9}$	$\underline{33.3_{0.0}}$	$29.2_{5.9}$	$29.2_{5.9}$	$29.2_{5.9}$	$41.1_{0.8}$	$40.7_{0.9}$
Gemini-2.5-pro	$38.9_{4.8}$	$41.7_{0.0}$	$30.6_{4.8}$	$38.9_{9.6}$	$\underline{47.2_{4.8}}$	$47.2_{4.8}$	$72.2_{12.7}$	$75.0_{8.3}$	$19.4_{4.8}$	$19.4_{4.8}$	$27.8_{4.8}$	$27.8_{4.8}$	$41.7_{8.3}$	$41.7_{8.3}$	$39.7_{3.8}$	$41.8_{3.1}$
Gemini-2.5-flash	$36.1_{17.3}$	$41.7_{22.0}$	$25.0_{8.3}$	$27.8_{4.8}$	$33.3_{0.0}$	$33.3_{0.0}$	$\underline{75.0_{8.3}}$	$72.2_{4.8}$	$30.6_{4.8}$	$27.8_{4.8}$	$22.2_{9.6}$	$19.4_{4.8}$	$38.9_{19.2}$	$36.1_{17.3}$	$37.3_{2.5}$	$37.7_{3.6}$
GPT-4o	$19.4_{4.8}$	$13.9_{4.8}$	$22.2_{4.8}$	$22.2_{4.8}$	$25.0_{8.3}$	$27.8_{4.8}$	$52.8_{9.6}$	$52.8_{9.6}$	$11.1_{4.8}$	$11.1_{4.8}$	$22.2_{4.8}$	$13.9_{4.8}$	$13.9_{4.8}$	$13.9_{4.8}$	$23.8_{1.2}$	$22.3_{1.4}$
Grok-3	$44.4_{4.8}$	$41.7_{8.3}$	$27.8_{4.8}$	$33.3_{8.3}$	$30.6_{9.6}$	$33.3_{8.3}$	$63.9_{4.8}$	$66.7_{8.3}$	$\underline{38.9_{4.8}}$	$\underline{38.9_{4.8}}$	$19.4_{4.8}$	$25.0_{8.3}$	$47.2_{4.8}$	$47.2_{4.8}$	$38.9_{1.8}$	$41.9_{2.6}$
Grok-4	$41.7_{0.0}$	$\mathbf{50.0_{0.0}}$	$25.0_{0.0}$	$33.3_{0.0}$	$\mathbf{50.0_{0.0}}$	$50.0_{0.0}$	$75.0_{0.0}$	$75.0_{0.0}$	$37.5_{5.9}$	$33.3_{0.0}$	$25.0_{0.0}$	$25.0_{0.0}$	$\mathbf{62.5_{5.9}}$	$\mathbf{62.5_{5.9}}$	$\underline{45.2_{0.0}}$	$\underline{48.3_{0.9}}$
o3	$\underline{50.0_{0.0}}$	$\mathbf{50.0_{0.0}}$	$33.3_{8.3}$	$41.7_{8.3}$	$44.4_{4.8}$	$50.0_{8.3}$	$72.2_{4.8}$	$72.2_{4.8}$	$\mathbf{41.7_{8.3}}$	$\mathbf{41.7_{8.3}}$	$\underline{33.3_{8.3}}$	$27.8_{4.8}$	$\underline{61.1_{4.8}}$	$\underline{61.1_{4.8}}$	$\mathbf{48.0_{1.8}}$	$\mathbf{49.7_{0.7}}$
o3 mini	$41.7_{8.3}$	$41.7_{8.3}$	$25.0_{0.0}$	$36.1_{4.8}$	$25.0_{8.3}$	$25.0_{8.3}$	$\mathbf{80.6_{4.8}}$	$\mathbf{80.6_{4.8}}$	$33.3_{0.0}$	$33.3_{0.0}$	$22.2_{4.8}$	$19.4_{4.8}$	$38.9_{9.6}$	$38.9_{9.6}$	$38.1_{2.4}$	$39.4_{2.0}$
o4 mini	$47.2_{4.8}$	$\mathbf{50.0_{0.0}}$	$33.3_{0.0}$	$41.7_{0.0}$	$36.1_{4.8}$	$36.1_{4.8}$	$72.2_{12.7}$	$72.2_{12.7}$	$33.3_{8.3}$	$\mathbf{41.7_{8.3}}$	$\underline{33.3_{8.3}}$	$27.8_{9.6}$	$47.2_{12.7}$	$47.2_{12.7}$	$43.3_{3.4}$	$46.4_{2.2}$
GPT-5 mini	$41.7_{8.3}$	$41.7_{0.0}$	$19.4_{4.8}$	$22.2_{9.6}$	$36.1_{12.7}$	$36.1_{12.7}$	$72.2_{4.8}$	$77.8_{4.8}$	$19.4_{9.6}$	$27.8_{9.6}$	$30.6_{4.8}$	$\mathbf{36.1_{4.8}}$	$41.7_{8.3}$	$41.7_{8.3}$	$37.3_{2.5}$	$41.3_{2.5}$
open-source models																
Qwen2.5−72B	$13.9_{4.8}$	$8.3_{8.3}$	$16.7_{0.0}$	$16.7_{0.0}$	$16.7_{0.0}$	$19.4_{4.8}$	$30.6_{9.6}$	$30.6_{9.6}$	$13.9_{4.8}$	$11.1_{4.8}$	$25.0_{0.0}$	$16.7_{0.0}$	$16.7_{0.0}$	$16.7_{0.0}$	$19.0_{2.4}$	$17.5_{3.0}$
Qwen3−32B	$19.4_{12.7}$	$16.7_{14.4}$	$33.3_{8.3}$	$36.1_{4.8}$	$25.0_{8.3}$	$38.9_{12.7}$	$52.8_{4.8}$	$52.8_{4.8}$	$27.8_{4.8}$	$30.6_{4.8}$	$22.2_{4.8}$	$22.2_{4.8}$	$27.8_{4.8}$	$27.8_{4.8}$	$29.8_{0.0}$	$31.7_{1.2}$
Qwen3−235B	$27.8_{4.8}$	$27.8_{4.8}$	$27.8_{4.8}$	$30.6_{4.8}$	$25.0_{0.0}$	$27.8_{4.8}$	$50.0_{0.0}$	$50.0_{0.0}$	$36.1_{9.6}$	$36.1_{9.6}$	$30.6_{4.8}$	$30.6_{4.8}$	$44.4_{4.8}$	$47.2_{4.8}$	$34.5_{2.4}$	$35.9_{2.5}$
QwQ-32B	$45.8_{5.9}$	$\underline{45.8_{5.9}}$	$25.0_{0.0}$	$29.2_{5.9}$	$37.5_{5.9}$	$37.5_{5.9}$	$\underline{75.0_{0.0}}$	$\underline{79.2_{5.9}}$	$25.0_{0.0}$	$25.0_{0.0}$	$\mathbf{37.5_{5.9}}$	$\underline{33.3_{0.0}}$	$50.0_{0.0}$	$50.0_{0.0}$	$42.3_{0.8}$	$42.6_{1.8}$
DeepSeek-V3	$33.3_{16.7}$	$27.8_{17.3}$	$36.1_{4.8}$	$36.1_{4.8}$	$33.3_{8.3}$	$36.1_{9.6}$	$55.6_{4.8}$	$55.6_{4.8}$	$36.1_{17.3}$	$36.1_{17.3}$	$27.8_{4.8}$	$22.2_{9.6}$	$36.1_{4.8}$	$36.1_{4.8}$	$36.9_{2.1}$	$35.4_{1.1}$
DeepSeek-V3.1	$\mathbf{52.8_{9.6}}$	$44.4_{9.6}$	$30.6_{4.8}$	$30.6_{4.8}$	$27.8_{4.8}$	$30.6_{12.7}$	$58.3_{14.4}$	$58.3_{14.4}$	$27.8_{4.8}$	$27.8_{4.8}$	$30.6_{4.8}$	$22.2_{4.8}$	$44.4_{4.8}$	$38.9_{9.6}$	$38.9_{2.5}$	$36.1_{3.8}$
DeepSeek-R1	$41.7_{11.8}$	$\mathbf{50.0_{0.0}}$	$25.0_{0.0}$	$37.5_{5.9}$	$33.3_{23.6}$	$41.7_{11.8}$	$62.5_{5.9}$	$62.5_{5.9}$	$33.3_{0.0}$	$33.3_{0.0}$	$20.8_{17.7}$	$20.8_{17.7}$	$61.0_{3.7}$	$61.0_{3.7}$	$39.7_{7.3}$	$45.2_{3.1}$
Gemma-3−27B-it	$16.7_{0.0}$	$8.3_{0.0}$	$27.8_{4.8}$	$27.8_{4.8}$	$19.4_{4.8}$	$19.4_{4.8}$	$44.4_{4.8}$	$44.4_{4.8}$	$16.7_{14.4}$	$16.7_{14.4}$	$13.9_{4.8}$	$11.1_{4.8}$	$36.1_{4.8}$	$36.1_{4.8}$	$25.0_{1.2}$	$24.6_{0.1}$
Llama-3−405b-I	$13.9_{4.8}$	$5.6_{4.8}$	$25.0_{8.3}$	$25.0_{8.3}$	$19.4_{15.7}$	$22.2_{12.7}$	$30.6_{4.8}$	$30.6_{4.8}$	$13.9_{4.8}$	$11.1_{4.8}$	$19.4_{4.8}$	$11.1_{4.8}$	$16.7_{8.3}$	$16.7_{8.3}$	$19.8_{2.7}$	$17.1_{3.3}$
Llama-3.3−70b-I	$8.3_{0.0}$	$0.0_{0.0}$	$16.7_{8.3}$	$16.7_{8.3}$	$22.2_{9.6}$	$16.7_{14.4}$	$27.8_{9.6}$	$30.6_{4.8}$	$16.7_{0.0}$	$16.7_{0.0}$	$25.0_{8.3}$	$16.7_{8.3}$	$33.3_{8.3}$	$36.1_{4.8}$	$21.4_{2.4}$	$18.9_{3.0}$
GPT-oss-20B	$\underline{50.0_{0.0}}$	$\mathbf{50.0_{0.0}}$	$25.0_{0.0}$	$38.9_{4.8}$	$\mathbf{50.0_{8.3}}$	$\mathbf{61.1_{9.6}}$	$61.1_{4.8}$	$61.1_{4.8}$	$25.0_{8.3}$	$33.3_{8.3}$	$\underline{33.3_{0.0}}$	$27.8_{4.8}$	$36.1_{9.6}$	$36.1_{9.6}$	$40.1_{2.7}$	$44.2_{2.6}$
GPT-oss-120B	$\mathbf{52.8_{4.8}}$	$\mathbf{50.0_{0.0}}$	$\mathbf{47.2_{9.6}}$	$\mathbf{58.3_{8.3}}$	$33.3_{8.3}$	$44.4_{4.8}$	$55.6_{21.0}$	$66.7_{16.7}$	$19.4_{4.8}$	$25.0_{0.0}$	$\underline{33.3_{0.0}}$	$25.0_{0.0}$	$38.9_{12.7}$	$38.9_{12.7}$	$40.1_{4.8}$	$44.4_{3.7}$
Llama-4-Scout	$27.8_{4.8}$	$19.4_{4.8}$	$36.1_{4.8}$	$38.9_{4.8}$	$33.3_{8.3}$	$36.1_{4.8}$	$55.6_{9.6}$	$55.6_{9.6}$	$36.1_{9.6}$	$30.6_{4.8}$	$30.6_{4.8}$	$25.0_{0.0}$	$52.8_{4.8}$	$52.8_{4.8}$	$38.9_{5.0}$	$36.4_{2.5}$
avg. (task)	$33.8_{15.2}$	$31.6_{18.2}$	$28.6_{8.4}$	$32.0_{10.4}$	$31.9_{11.5}$	$34.9_{13.2}$	$58.0_{16.4}$	$59.1_{16.4}$	$27.5_{11.2}$	$28.2_{10.8}$	$27.1_{7.9}$	$23.3_{8.2}$	$38.6_{14.5}$	$41.0_{15.7}$		

[a]Bold indicates the best score per column, underlined indicates the second-best (excluding ties). Our experiments employ xVerify-0.5B−I. This is the result for the balanced benchmark dataset (QCBench-84), with 95% confidence interval.

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Table 5. Approximation and Exact Matching (by xVerify) Accuracies (%, ↑) of Models Across 7 Chemistry Subfields, with Columns for Approximate (Appr.) and xVerify Results[a]

Model	Analytical		Biochemistry		General		Inorganic		Physical		Polymer		Quantum		avg. (perf.)	
	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify	appr.	xVerify
closed-source models																
Claude-3.5-Sonnet	$18.7_{2.3}$	$22.7_{4.6}$	$22.7_{6.1}$	$24.0_{4.0}$	$20.8_{3.6}$	$20.8_{3.6}$	$42.0_{4.5}$	$44.2_{3.3}$	$31.4_{0.8}$	$29.4_{0.5}$	<u>$33.3_{14.4}$</u>	$27.8_{9.6}$	$23.1_{6.8}$	$24.8_{5.9}$	$27.4_{4.0}$	$27.7_{3.5}$
Claude-4-Sonnet	$26.0_{2.8}$	$32.0_{5.7}$	$42.0_{2.8}$	$42.0_{2.8}$	$46.9_{4.4}$	<u>$53.1_{4.4}$</u>	$55.4_{1.5}$	$53.3_{1.5}$	$41.4_{0.4}$	$38.5_{0.0}$	<u>$33.3_{0.0}$</u>	$29.2_{5.9}$	$38.5_{0.0}$	$38.5_{0.0}$	$40.5_{0.4}$	$40.9_{0.8}$
Gemini-2.5-pro	$30.7_{2.3}$	$36.0_{0.0}$	$30.7_{4.6}$	$33.3_{2.3}$	<u>$47.9_{3.6}$</u>	$47.9_{3.6}$	$55.8_{3.3}$	$58.0_{3.3}$	<u>$50.6_{1.3}$</u>	<u>$46.2_{2.2}$</u>	$27.8_{4.8}$	$27.8_{4.8}$	$42.7_{3.0}$	$43.6_{2.6}$	$40.9_{2.9}$	$41.8_{1.9}$
Gemini-2.5-flash	$25.3_{12.2}$	$25.3_{12.2}$	$33.3_{2.3}$	$32.0_{4.0}$	$43.8_{0.0}$	$43.8_{0.0}$	$57.2_{3.3}$	$54.3_{2.2}$	$33.0_{1.7}$	$30.5_{0.9}$	$22.2_{9.6}$	$19.4_{4.8}$	$30.8_{8.9}$	$30.8_{6.8}$	$35.1_{1.7}$	$33.7_{1.6}$
GPT-4o	$16.0_{6.9}$	$13.3_{6.1}$	$22.7_{2.3}$	$24.0_{0.0}$	$20.8_{70.2}$	$22.9_{3.6}$	$48.6_{1.3}$	$44.2_{2.5}$	$22.8_{2.6}$	$21.4_{1.9}$	$22.2_{4.8}$	$13.9_{4.8}$	$23.9_{3.9}$	$24.8_{3.0}$	$25.3_{1.4}$	$23.5_{1.4}$
Grok-3	$33.3_{2.3}$	$36.0_{4.0}$	$30.7_{6.1}$	$37.3_{4.6}$	$41.7_{7.2}$	$43.8_{6.2}$	$59.4_{4.5}$	$60.0_{4.3}$	$46.0_{3.5}$	<u>$46.2_{3.8}$</u>	$19.4_{4.8}$	$25.0_{8.3}$	$43.6_{5.1}$	$47.0_{3.9}$	$39.2_{2.6}$	$42.3_{3.1}$
Grok-4	$28.0_{5.7}$	**$46.0_{2.8}$**	$36.0_{5.7}$	$40.0_{5.7}$	**$50.0_{0.0}$**	$50.0_{0.0}$	$63.0_{3.1}$	$62.0_{1.5}$	**$52.7_{1.9}$**	**$51.2_{2.3}$**	$25.0_{0.0}$	$25.0_{0.0}$	$48.7_{3.6}$	$48.7_{3.6}$	<u>$43.3_{0.7}$</u>	**$46.1_{0.8}$**
o3	<u>$38.7_{2.3}$</u>	$33.3_{2.3}$	<u>$42.7_{2.3}$</u>	<u>$45.3_{4.6}$</u>	**$50.0_{0.0}$**	**$54.2_{3.6}$**	**$64.5_{1.3}$**	<u>$63.0_{2.2}$</u>	$47.2_{3.4}$	$45.1_{3.9}$	<u>$33.3_{8.3}$</u>	$27.8_{4.8}$	<u>$51.3_{2.6}$</u>	<u>$51.3_{2.6}$</u>	**$46.8_{1.8}$**	<u>$45.7_{1.0}$</u>
o3 mini	$33.3_{6.1}$	$29.3_{6.1}$	$33.3_{4.6}$	$42.7_{6.1}$	$22.9_{9.5}$	$22.9_{9.5}$	<u>$63.8_{1.3}$</u>	**$63.8_{2.5}$**	$39.8_{1.7}$	$38.1_{1.6}$	$22.2_{4.8}$	$19.4_{4.8}$	$43.6_{5.1}$	$44.4_{6.5}$	$37.0_{2.6}$	$37.2_{1.9}$
o4 mini	$34.7_{6.1}$	$30.7_{2.3}$	$37.3_{4.6}$	$42.7_{6.1}$	$35.4_{3.6}$	$35.4_{3.6}$	**$64.5_{4.5}$**	**$63.8_{7.0}$**	$40.5_{0.6}$	$39.4_{0.8}$	<u>$33.3_{8.3}$</u>	$27.8_{9.6}$	$41.9_{3.9}$	$44.4_{3.9}$	$41.1_{2.8}$	$40.6_{2.3}$
GPT-5 mini	$30.7_{2.3}$	$29.3_{2.3}$	$25.3_{6.1}$	$30.7_{8.3}$	$33.3_{9.5}$	$33.3_{9.5}$	$58.7_{2.2}$	$60.1_{1.3}$	$39.8_{2.2}$	$39.9_{2.0}$	$30.6_{4.8}$	**$36.1_{4.8}$**	$29.9_{3.9}$	$34.2_{3.0}$	$35.5_{2.4}$	$37.7_{3.4}$
Open-source Models																
Qwen2.5−72B	$13.3_{4.6}$	$6.7_{6.1}$	$18.7_{2.3}$	$18.7_{2.3}$	$12.5_{0.0}$	$14.6_{3.6}$	$37.0_{3.8}$	$33.3_{2.5}$	$21.2_{0.3}$	$19.4_{0.3}$	$25.0_{0.0}$	$16.7_{0.0}$	$22.2_{3.0}$	$21.4_{1.5}$	$21.4_{1.2}$	$18.7_{1.5}$
Qwen3−32B	$16.8_{8.0}$	$13.3_{6.1}$	$30.7_{2.3}$	$32.0_{4.0}$	$22.9_{3.6}$	$33.3_{7.2}$	$47.1_{1.3}$	$46.4_{2.5}$	$28.9_{2.8}$	$28.3_{1.6}$	$22.2_{4.8}$	$22.2_{4.8}$	$27.4_{3.0}$	$27.4_{3.0}$	$27.9_{1.8}$	$29.0_{1.4}$
Qwen3−235B	$18.7_{2.3}$	$18.7_{2.3}$	$29.3_{6.1}$	$30.7_{6.1}$	$25.0_{0.0}$	$27.1_{3.6}$	$50.7_{4.5}$	$46.4_{4.5}$	$35.8_{1.6}$	$33.2_{1.4}$	$30.6_{4.8}$	$30.6_{4.8}$	$32.5_{1.5}$	$35.0_{3.9}$	$31.8_{1.0}$	$31.6_{0.9}$
QwQ-32B	$32.0_{0.0}$	<u>$40.0_{0.0}$</u>	$28.0_{0.0}$	$30.0_{2.8}$	$37.5_{8.8}$	$37.5_{8.8}$	$58.7_{3.1}$	$58.7_{6.1}$	$46.0_{0.8}$	$44.1_{1.1}$	**$37.5_{5.9}$**	<u>$33.3_{0.0}$</u>	$48.7_{3.6}$	$48.7_{3.6}$	$41.2_{0.4}$	$41.8_{0.4}$
DeepSeek-V3	$24.0_{6.9}$	$24.0_{10.6}$	$29.3_{4.6}$	$29.3_{4.6}$	$29.2_{7.2}$	$31.2_{6.2}$	$49.3_{5.0}$	$44.9_{5.0}$	$33.7_{1.1}$	$31.2_{0.3}$	$27.8_{4.8}$	$22.2_{9.6}$	$31.6_{1.5}$	$31.6_{1.5}$	$32.1_{0.9}$	$30.7_{1.9}$
DeepSeek-V3.1	$36.0_{6.9}$	$34.7_{4.6}$	$33.3_{4.6}$	$34.7_{6.1}$	$33.3_{7.2}$	$35.4_{7.2}$	$52.9_{7.0}$	$51.4_{8.2}$	$36.2_{0.6}$	$32.8_{1.1}$	$30.6_{4.8}$	$22.2_{4.8}$	$35.0_{5.3}$	$33.3_{6.8}$	$36.8_{1.8}$	$34.9_{1.6}$
DeepSeek-R1	$34.0_{8.5}$	$46.0_{2.8}$	$30.0_{2.8}$	$36.0_{11.3}$	$34.4_{13.3}$	$43.8_{0.0}$	$56.5_{6.1}$	$58.7_{0.0}$	$40.6_{1.5}$	$39.8_{1.9}$	$20.8_{17.7}$	$20.8_{17.7}$	**$52.6_{1.8}$**	**$53.8_{3.6}$**	$38.4_{6.1}$	$42.7_{0.5}$
Gemma-3−27B-it	$12.0_{0.0}$	$4.0_{0.0}$	$17.3_{2.3}$	$17.3_{2.3}$	$14.6_{3.6}$	$14.6_{3.6}$	$34.1_{2.5}$	$31.9_{1.3}$	$20.5_{1.6}$	$20.3_{1.4}$	$13.9_{4.8}$	$11.1_{4.8}$	$22.2_{7.4}$	$26.5_{8.2}$	$19.2_{1.0}$	$18.0_{0.8}$
Llama-3−405b-I	$13.3_{6.1}$	$8.0_{4.0}$	$17.3_{2.3}$	$17.3_{2.3}$	$14.6_{9.5}$	$16.7_{9.5}$	$26.1_{5.8}$	$27.5_{4.5}$	$18.7_{3.2}$	$18.9_{2.5}$	$19.4_{4.8}$	$11.1_{4.8}$	$22.2_{1.5}$	$23.9_{3.0}$	$18.8_{2.8}$	$17.8_{3.3}$
Llama-3.3−70b-I	$10.7_{2.3}$	$4.0_{0.0}$	$16.0_{4.0}$	$16.0_{4.0}$	$16.7_{7.2}$	$12.5_{10.8}$	$31.2_{4.5}$	$29.7_{3.3}$	$20.7_{1.7}$	$19.6_{2.2}$	$25.0_{8.3}$	$16.7_{8.3}$	$21.4_{3.0}$	$23.9_{3.9}$	$20.2_{1.7}$	$17.5_{2.3}$
GPT-oss-20B	$32.0_{4.0}$	<u>$40.0_{0.0}$</u>	$29.3_{2.3}$	$38.7_{4.6}$	$41.7_{9.5}$	$50.0_{10.8}$	$56.5_{2.2}$	$58.0_{1.3}$	$37.4_{3.2}$	$38.0_{4.0}$	<u>$33.3_{0.0}$</u>	$27.8_{4.8}$	$37.6_{1.5}$	$40.2_{1.5}$	$38.3_{1.7}$	$41.8_{1.3}$
GPT-oss-120B	**$40.0_{0.0}$**	$32.0_{0.0}$	**$54.7_{6.1}$**	**$64.0_{0.0}$**	$25.0_{6.2}$	$33.3_{3.6}$	$54.3_{7.8}$	$59.4_{4.5}$	$35.1_{2.2}$	$39.4_{1.6}$	<u>$33.3_{0.0}$</u>	$25.0_{0.0}$	$38.5_{7.7}$	$40.2_{7.8}$	$40.1_{2.2}$	$41.9_{1.8}$
Llama-4-Scout	$16.0_{4.0}$	$10.7_{2.3}$	$29.3_{2.3}$	$30.7_{2.3}$	$25.0_{6.2}$	$27.1_{3.6}$	$48.6_{3.3}$	$45.7_{5.8}$	$34.2_{1.4}$	$31.0_{0.9}$	$30.6_{4.8}$	$25.0_{0.0}$	$46.2_{0.0}$	$45.3_{1.5}$	$32.8_{2.0}$	$30.8_{0.6}$
avg. (task)	$25.3_{10.3}$	$24.8_{13.2}$	$29.8_{9.5}$	$32.7_{11.4}$	$30.4_{12.8}$	$32.8_{13.6}$	$51.1_{11.2}$	$50.4_{11.8}$	$35.0_{9.6}$	$33.7_{9.4}$	$27.1_{7.9}$	$23.3_{8.2}$	$35.0_{10.5}$	$36.2_{10.3}$		

[a]Bold indicates the best score per column, underlined indicates the second-best (excluding ties). Our experiments employ xVerify-0.5B−I. This is the result for the full benchmark dataset (QCBench-350), with 95% confidence interval.

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other leading models like **Grok-4** and **QwQ-32B** maintain their strong positions with only minor fluctuations in their average approximate scores (typically within 1−2 percentage points). This stability suggests that the balanced subset is a reliable proxy for general model capability.

Challenging Subfields. From the average task performance on the full benchmark data set QCBench-350 (Table 5), we observe a clear hierarchy of difficulty across the chemistry subfields for current large language models. *Analytical Chemistry* and *Polymer Chemistry* pose the most significant challenges, with the lowest average approximate matching scores of 25.3 and 27.1%, respectively. The difficulty in *Analytical Chemistry* likely stems from its requirement for multistep reasoning, precise stoichiometric calculations, and an implicit understanding of experimental procedures and data interpretation (e.g., titrations, spectroscopy). Similarly, *Polymer Chemistry* problems often involve complex concepts such as molecular weight distributions, polymerization kinetics, and the manipulation of intricate structural formulas, which appear to be outside the core competency of most generalized models. In stark contrast, models demonstrate the highest proficiency in *Inorganic Chemistry*, achieving a notable average approximate score of 51.1%. This suggests that the principles of inorganic chemistry, such as periodic trends, coordination compounds, and basic reaction balancing, might be better represented in the models' vast training corpora. *Physical Chemistry* and *Quantum Chemistry* occupy a middle ground in terms of difficulty, with average scores of 35.0%. These fields demand a strong grasp of mathematical formalism and physical laws, indicating that while models have some capability in these areas, there is substantial room for improvement. This disparity underscores a critical insight: current LLMs are not uniformly proficient across all scientific domains, and their performance is heavily influenced by the nature and complexity of the calculations required.

Closed-Source vs Open-Source Models. Our results reveal a nuanced landscape when comparing closed-source and open-source models. While on average, the top-tier closed-source models exhibit a performance advantage, the gap is not absolute, and leading open-source models demonstrate state-of-the-art capabilities in specific subfields. Among all evaluated models, the closed-source model **o3** emerges as the overall top performer, achieving the highest average approximate score of 46.8%. Its strength is not confined to a single area; it achieves the best performance in both *General Chemistry* (50.0%) and *Inorganic Chemistry* (64.5%), and secures the second-best position in highly challenging fields like *Biochemistry* and *Quantum Chemistry*. Following closely is **Grok-4**, another powerful closed-source model with an average score of 43.3%. **Grok-4** particularly excels in the mathematically intensive field of *Physical Chemistry*, where it obtains the highest score of 52.7%, and it shares the top rank in *General Chemistry*, showcasing its robust and versatile reasoning abilities.

The open-source community has produced highly competitive models that challenge the dominance of their closed-source counterparts in specialized domains. **QwQ-32B** stands out as the leading open-source model with a strong average approximate accuracy of 41.2%. Impressively, it achieves the absolute top score in *Polymer Chemistry* (37.5%), one of the most difficult subfields for all models. This indicates a high degree of specialization. Furthermore, **GPT-oss-120B** delivers an exceptional performance, nearly matching the top open-source model with an average score of 40.1%. Its capabilities are particularly noteworthy as it achieves the highest scores across all models in both *Analytical Chemistry* (40.0%) and *Biochemistry* (54.7%), demonstrating that targeted open-source efforts can lead to state-of-the-art results. Similarly, **DeepSeek-R1** proves its mettle by obtaining the top score in *Quantum Chemistry* (52.6%). At the other end of the spectrum, models such as **Llama-3−405b-I** (18.8%) and **Gemma-3−27B-it** (19.2%) consistently rank among the weakest performers. Their low scores across nearly all subfields highlight the benchmark's effectiveness in differentiating model capabilities and underscore the significant challenges that quantitative chemistry poses to nonspecialized large language models.

Difficulty Levels vs Performance. To investigate how model performance scales with task complexity, we first established a baseline difficulty categorization. This initial heuristic, detailed in the Supporting Information (Section: Data Preparation), stratifies problems into easy, medium, and difficult tiers based on their input length. While serving as a useful starting point, this length-based proxy for complexity revealed a series of counterintuitive and intriguing performance dynamics, as shown in the following analysis.

Figures 4 and 5 illustrate the performance of representative models across three difficulty tiers (easy, medium, and difficult) by length, evaluated using Approximate Matching and the stricter xVerify, respectively. Our analysis reveals three distinct model archetypes.

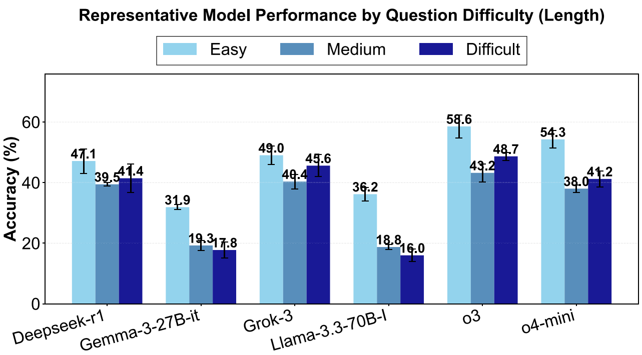

Figure 4. Accuracies on difficulty levels (Approximate Matching on QCBench-350).

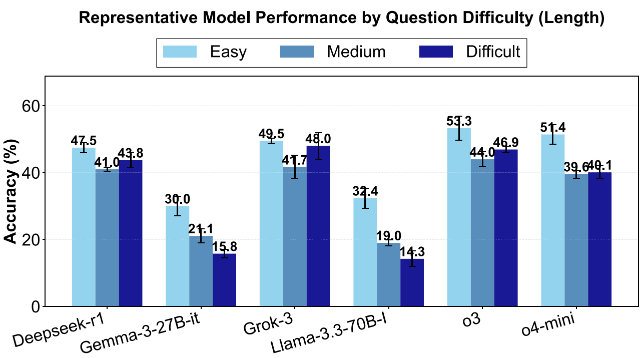

Figure 5. Accuracies on difficulty levels (xVerify on QCBench-350).

Robust and Counterintuitive Models. Elite models like **Grok-3** and **DeepSeek-R1** exhibit top-tier performance but display a counterintuitive trend: their accuracy on medium and difficult questions sometimes surpasses their performance on easy ones. While this may indicate advanced reasoning capabilities, a deeper analysis suggests two contributing factors.

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First, our length-based difficulty heuristic may create artifacts where shorter easy questions lack sufficient context. Second, we observe a clear pattern of overthinking: these models often apply unnecessarily complex reasoning to simple problems, leading them to infer nonexistent complexities and ultimately arrive at incorrect solutions.

Strong but Nuanced Models. Unlike a conventional model that shows a simple decline, some models like **o3** and **o4 mini** exhibit a more complex profile. Both models achieve their highest accuracy on easy questions, as expected, with **o3** reaching an impressive 58.6%. However, both models also perform better on difficult questions than on medium ones. For example, **o3**'s accuracy is 48.7% on difficult tasks compared to 43.2% on medium tasks. This indicates that while these models are well-rounded, they possess strong capabilities for handling complex problems, outperforming their results on intermediate-difficulty questions.

Difficulty-Sensitive Models. Some models show a high sensitivity to question difficulty. They exhibit a clear and significant drop-off in performance as the complexity rises. For instance, **Gemma-3−27B-it**'s accuracy falls from 31.9% on easy questions to just 17.8% on difficult ones. Similarly, **Llama-3.3−70B−I** sees its performance halved, from 36.2% on easy ones to 16.0% on difficult ones. This steep, monotonic decline is characteristic of models that may lack the specialized reasoning capacity required for more complex, multistep chemical problems.

Recalibration of Difficulty Levels by Qwen3-Max. Recognizing that heuristics such as problem length provide only a superficial measure of difficulty, we established a more robust and semantically meaningful categorization for our analysis. We used a state-of-the-art large language model, **Qwen3-Max**, to re-evaluate and assign a difficulty level (easy, medium, or difficult) to each question in the QCBench data set. To ensure impartiality and prevent any potential bias from data leakage or circular evaluation, Qwen3-Max was used exclusively for this annotation task and was not included in our set of evaluated models. The performance of representative models under this new, more nuanced difficulty scheme is presented in Figures 6 and 7.

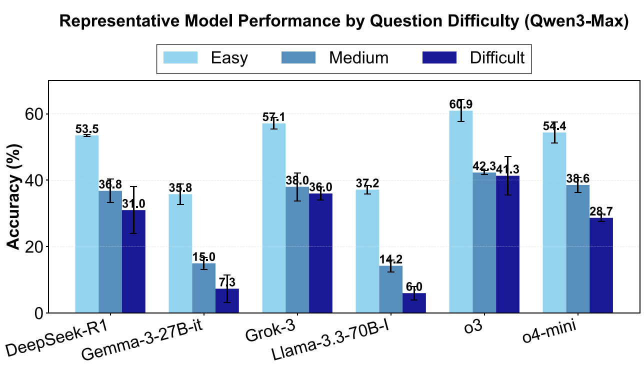

Figure 6. Accuracies on difficulty levels (approximate matching on QCBench-350 by Qwen3-Max).

Under the Qwen3-Max difficulty ratings, the previously observed counterintuitive trends in elite models have largely disappeared. These models now exhibit a more predictable performance curve: they achieve their highest accuracy on easy questions and show an expected decline as the semantic difficulty increases. For example, **Grok-3** and **o3** demonstrate

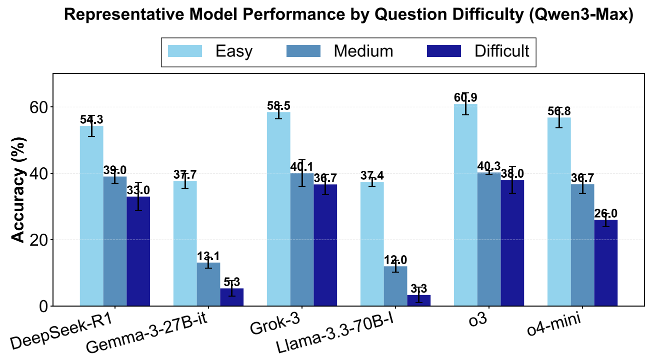

Figure 7. Accuracies on difficulty levels (xVerify on QCBench-350 by Qwen3-Max).

exceptional capability on easy tasks, with accuracies reaching 59.0 and 57.4%, respectively. Their performance then moderately decreases on medium (40.4 and 42.7%) and difficult (38.0% for both) questions. Similarly, **DeepSeek-R1** and **o4 mini** follow this pattern, with a monotonic decrease in accuracy from easy to difficult tasks (Table 6). This suggests that the previously unusual performance curves were likely an artifact of the superficial length-based difficulty metric rather than an inherent property of the models' reasoning (Table 7).

Table 6. Distribution of Question Difficulties Across Chemistry Subfields in QCBench (350 Questions) by Qwen3-Max

class	abbrev.	easy	medium	difficult
Analytical	AC	2	12	11
Biochemistry	BOC	11	10	3
General	GC	3	11	2
Inorganic	IC	15	25	6
Physical	PhC	70	97	20
Polymer	PoC	3	5	4
Quantum	QC	18	17	4
total		122	178	50

Table 7. Distribution of Question Difficulties Across Chemistry Subfields in QCBench-Balanced (84 Questions) by Qwen3-Max

class	abbrev.	easy	medium	difficult
Analytical	AC	2	4	6
Biochemistry	BOC	7	5	0
General	GC	3	8	1
Inorganic	IC	3	9	0
Physical	PhC	6	4	2
Polymer	PoC	3	5	4
Quantum	QC	7	3	2
total		31	38	15

Approximate Verification vs Exact Matching. An analysis of the benchmark's approximate accuracy (Appr.) and the exact matching enabled by xVerify reveals a crucial dynamic: for many advanced models, the verification process functions as a powerful self-correction and reasoning-enhancement tool. This is a significant shift from the notion of a verifier merely acting as a restrictive filter. The data in Table 5 shows that numerous top-performing models achieve a higher score under the stricter xVerify metric than with the more

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lenient approximate matching. This trend is exemplified by several leading models. For instance, the closed-source model **Grok-4** sees its average performance increase from 43.3% (Appr.) to 46.1% (xVerify). Even more striking are the gains observed in the open-source domain: **DeepSeek-R1**'s accuracy jumps from 38.4 to 42.7%, and **GPT-oss-120B** improves from 40.1 to 41.9%. This performance uplift indicates that these models possess a sophisticated internal reasoning capability. When prompted to verify their work, they can retrace their steps, identify initial fallacies—be they calculation errors or flawed logic—and converge on the correct, verifiable answer. The verification step is not just about formatting; it is an active problem-solving phase. For example, at the index 166 for computing the free energy difference giving the dissociation constant for the particular protein dimer, **Grok-4** gives the answer −34.2, where the groundtruth is 34.2. Although a direct numerical comparison would flag the answer as incorrect, xVerify's more sophisticated evaluation correctly identifies it as valid, demonstrating its ability to handle chemically meaningful equivalences. However, this self-correction capability is not universal and serves as a key differentiator. Less capable models, such as **Llama-3−405b-I** (18.8% Appr. vs 17.8% xVerify) and **Gemma-3−27B-it** (19.2 vs 18.0%), tend to see their scores decrease after verification. This suggests that their initial correct answers might be a result of lucky guesses or superficial pattern matching, and their underlying reasoning is not robust enough to withstand the scrutiny of a verification process. One typical example is the index 140 of the QCBench-350, where the answer is 3.548×10^{-27}. The predicted answer by **Gemma3−27B-it** is 3.54×10^{-24}, If we directly compare the difference, it is small because of the scale. Therefore, the gap between Appr. and xVerify is no longer an indicator of a verifier's limitation. Instead, a positive gap (where xVerify > Appr.) serves as a strong signal of a model's advanced, reliable reasoning and self-correction abilities, separating models that can genuinely solve a problem from those that can only approximate a plausible-sounding answer.

■ RANKING DECOUPLING ACROSS CHEMICAL BENCHMARKS

To contextualize our findings and validate the unique contribution of QCBench, we compared the model performance hierarchy with that of ChemBench,[9] a well-established, broader benchmark for chemical capabilities. Instead of finding consistent rankings, our analysis reveals a significant and insightful **ranking decoupling** between the two benchmarks, highlighting the specialized nature of the skills measured by QCBench. The comparative analysis for models present in both our full QCBench-350 data set (Table 5) and the ChemBench leaderboard is summarized in Table 8.

The results are striking. With the exception of the top-performing **o3** model, there is a dramatic divergence in rankings. Most notably, **Claude-3.5 Sonnet**, a top-3 model on ChemBench, plummets to 19th place on QCBench, with its absolute accuracy dropping from 62% to a mere 27.4%. Similarly, other highly ranked models on ChemBench, such as **GPT-4o** and **Llama-3.3−70B−I**, experience a severe drop in both rank and performance on our benchmark. This decoupling phenomenon strongly supports our central thesis. Performance on a broad-domain benchmark like ChemBench reflects a model's ability as a comprehensive chemical knowledge base, excelling at qualitative, factual, and simple predictive tasks. In contrast, QCBench is specifically designed to evaluate a model's capability as a *Chemical Computational Engine*, a skill that demands rigorous, multistep mathematical reasoning. The stark performance drop demonstrates that these two capabilities are not interchangeable. A model can possess vast chemical knowledge yet falter when required to perform complex calculations. Therefore, this analysis validates QCBench not as a mere alternative to existing benchmarks, but as an essential and orthogonal diagnostic tool. It successfully isolates and quantifies a core, yet often obscured, dimension of scientific intelligence, revealing critical weaknesses in quantitative reasoning that are masked by strong performance in other areas.

Table 8. Demonstration of Ranking Decoupling between QCBench (Quantitative Focus) and ChemBench (Broad Focus)[a]

	QCBench		ChemBench[9]	
model	rank	appr. acc.	rank	overall acc.
o1/o3	**first**	0.47	**first**	0.64
Claude-3.5 Sonnet	19th	0.27	**third**	0.62
GPT-4o	16th	0.35	fourth	0.61
Llama-3.3−70B-Instruct	22nd	0.19	seventh	0.52

[a]The stark drop in ranks highlights the difference between general chemical knowledge and quantitative reasoning ability.

■ CONCLUSION

In this work, we introduced QCBench, a benchmark meticulously designed to move beyond qualitative assessments and rigorously probe the quantitative reasoning of LLMs, which is a critical, yet underexplored facet of chemical intelligence. Our comprehensive evaluation using this framework has yielded several critical insights that challenge common assumptions in the field. We confirmed a profound gap between the linguistic fluency of LLMs and their computational accuracy, with domains like *Analytical* and *Polymer Chemistry* acting as key frontiers. Furthermore, our findings demonstrate that top-tier performance is not merely a function of model scale, but a nuanced interplay between specialized reasoning strengths and computational efficiency. Perhaps most significantly, our analysis of the verification gap reveals a crucial methodological insight: for elite models, strict verification can act as a reasoning enhancer rather than a penalty, suggesting that evaluation tools themselves must coevolve with the models they assess. Ultimately, QCBench offers more than a static leaderboard. It provides a dynamic diagnostic framework that pinpoints specific weaknesses and guides the targeted development of more scientifically robust and accurate AI. This work lays the groundwork for future research into domain-specific fine-tuning, advanced self-correction mechanisms, and a deeper understanding of emergent computational capabilities in scientific AI.

■ ASSOCIATED CONTENT

Data Availability Statement

All data is available at https://huggingface.co/datasets/jiaxie/QCBench. Code is available at https://github.com/jiaqingxie/QCBench. The API key should have access to the models.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.5c02033.

Data preparation, experimental setup, ablation studies including tolerance and thinking mode analysis, model performance results, and limitations for the QCBench quantitative chemistry benchmark (PDF)

AUTHOR INFORMATION

Corresponding Author

Yuqiang Li − *Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China*; orcid.org/0000-0001-6756-6154; Email: liyuqiang@pjlab.org.cn

Authors

Jiaqing Xie − *Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China*; orcid.org/0000-0001-7634-4457

Weida Wang − *Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China; Fudan University, Shanghai 200433, China*

Ben Gao − *Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China; Wuhan University, Wuhan, Hubei 430072, China*

Zhuo Yang − *Xidian University, Xi'an, Shaanxi 710126, China*

Haiyuan Wan − *Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China; Tsinghua University, Beijing 100084, China*

Shufei Zhang − *Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China*

Tianfan Fu − *Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China; Nanjing University, Nanjing, Jiangsu 210023, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jcim.5c02033

Author Contributions

$^{\nabla}$J.X., W.W. and B.G. contributed equally to this work. J.X. is responsible for data curation, drafting the manuscript, visualization, and computational experiments. W.W. is responsible for coding, drafting the manuscript, computational experiments, and visualization. B.G. is responsible for drafting the manuscript, and data curation. Z.Y. is responsible for drafting the manuscript. S.Z. is responsible for technical guidance. T.F. is responsible for paper proofreading. Y.L. is responsible for supporting the whole project.

Notes

The authors declare no competing financial interest.

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